

Swiss Institute of Banking and Finance



University of St.Gallen

2. Risk and Return

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3,130/4,130 Corporate Finance

Course Outline

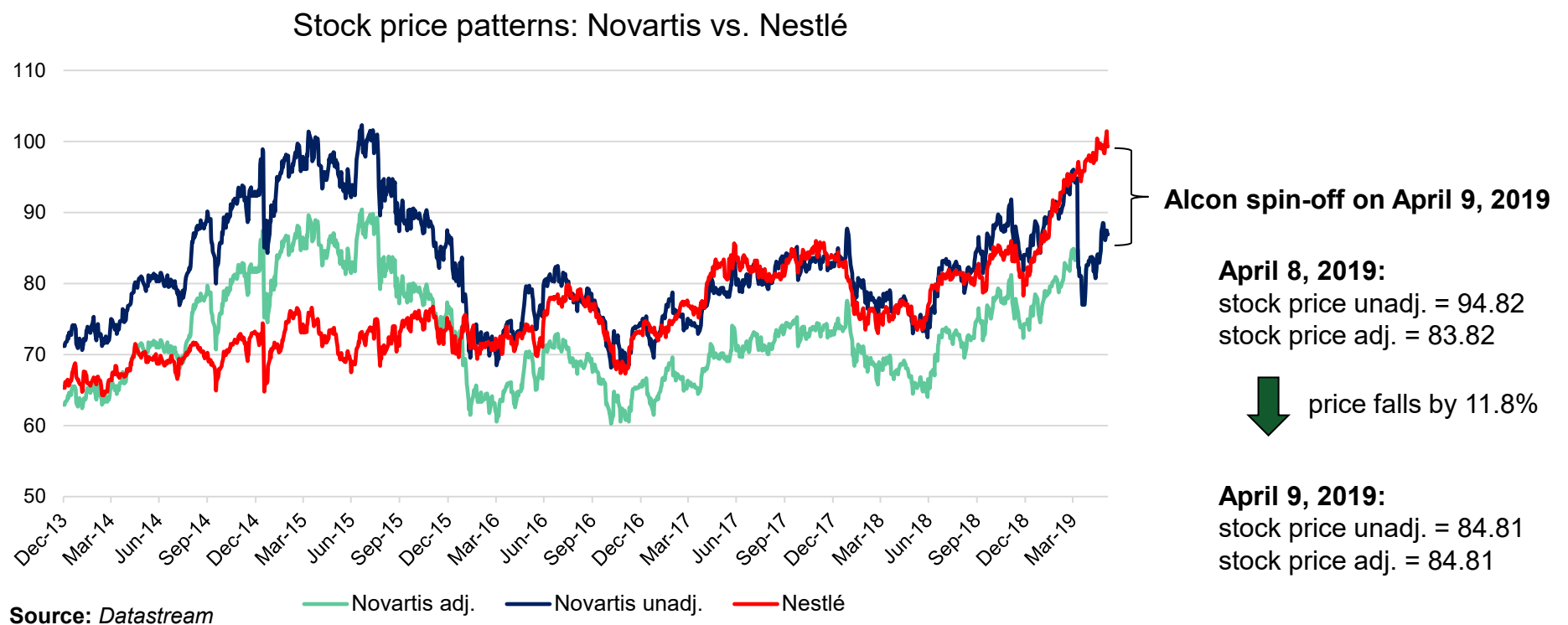
1. General Information
- 2. Risk and Return**
3. Portfolio Theory and CAPM
4. Corporate Valuation
5. Cost of Capital
6. Corporate Financing and the Issuance of Securities
7. Capital Structure
8. Risk Management
9. Conflicts of Interest and Corporate Governance
10. Market Efficiency and Behavioral (Corporate) Finance
11. Empirical Research in Corporate Finance

After this lecture you will have understood...

- Why **different investments** show **different returns**,
- what the **equity risk premium** is,
- why **different investments** show **varying degrees of risk**,
- how **risk** of individual securities and of a portfolio of securities can be **measured**,
- how **diversification** reduces risk (and the limits to diversification),
- and the **concept of “beta”**.

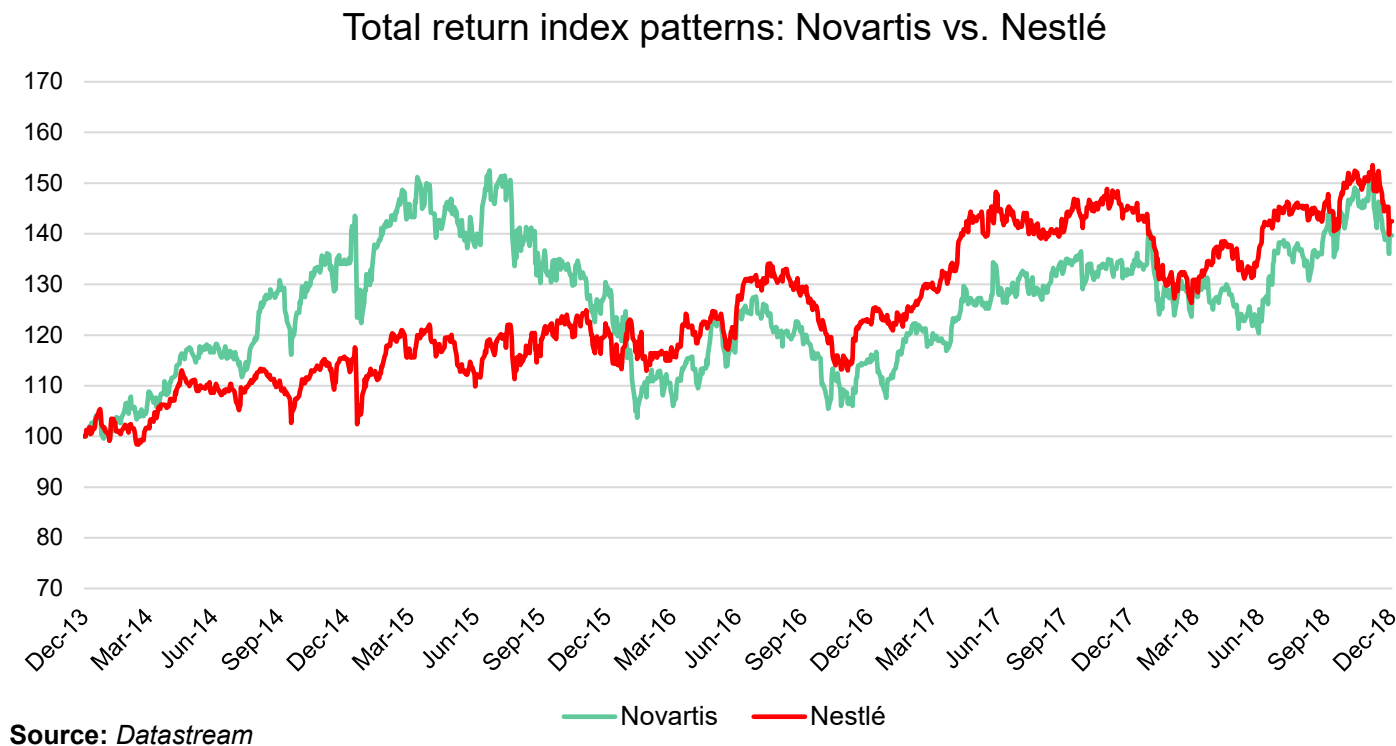
How do we compare the risk level of different companies?

- Which of the following two companies is **more risky** and why?



How do we compare the risk level of different companies? (II)

- Which of the following two companies is **more risky** and why?

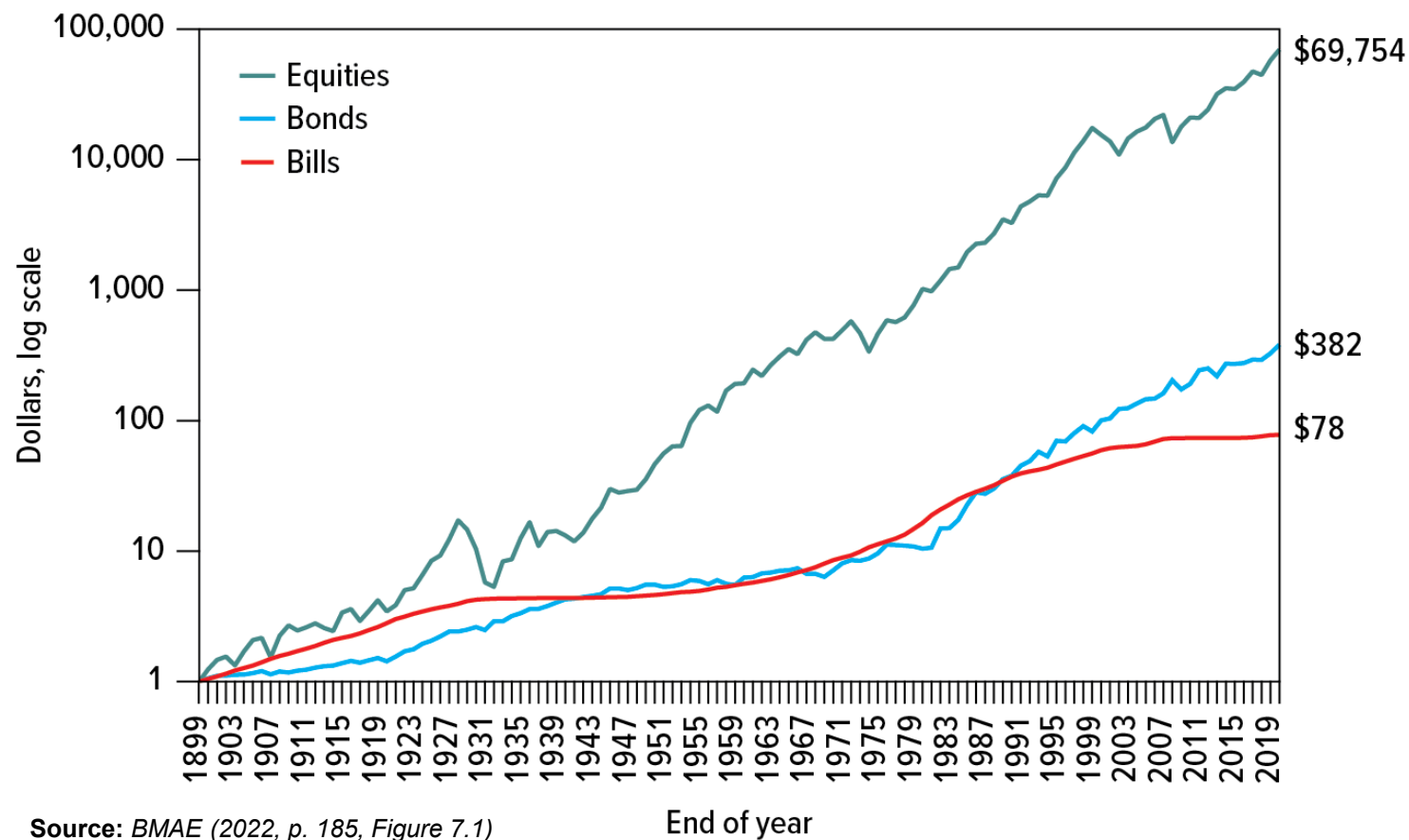


Different investments and their performance

- For simplicity, we start by analyzing three possible investments:
 1. A portfolio of Treasury Bills (“Bills”)
 2. A portfolio of U.S. government bonds (“Bonds”)
 3. A portfolio of U.S. common stocks (“Stocks”)

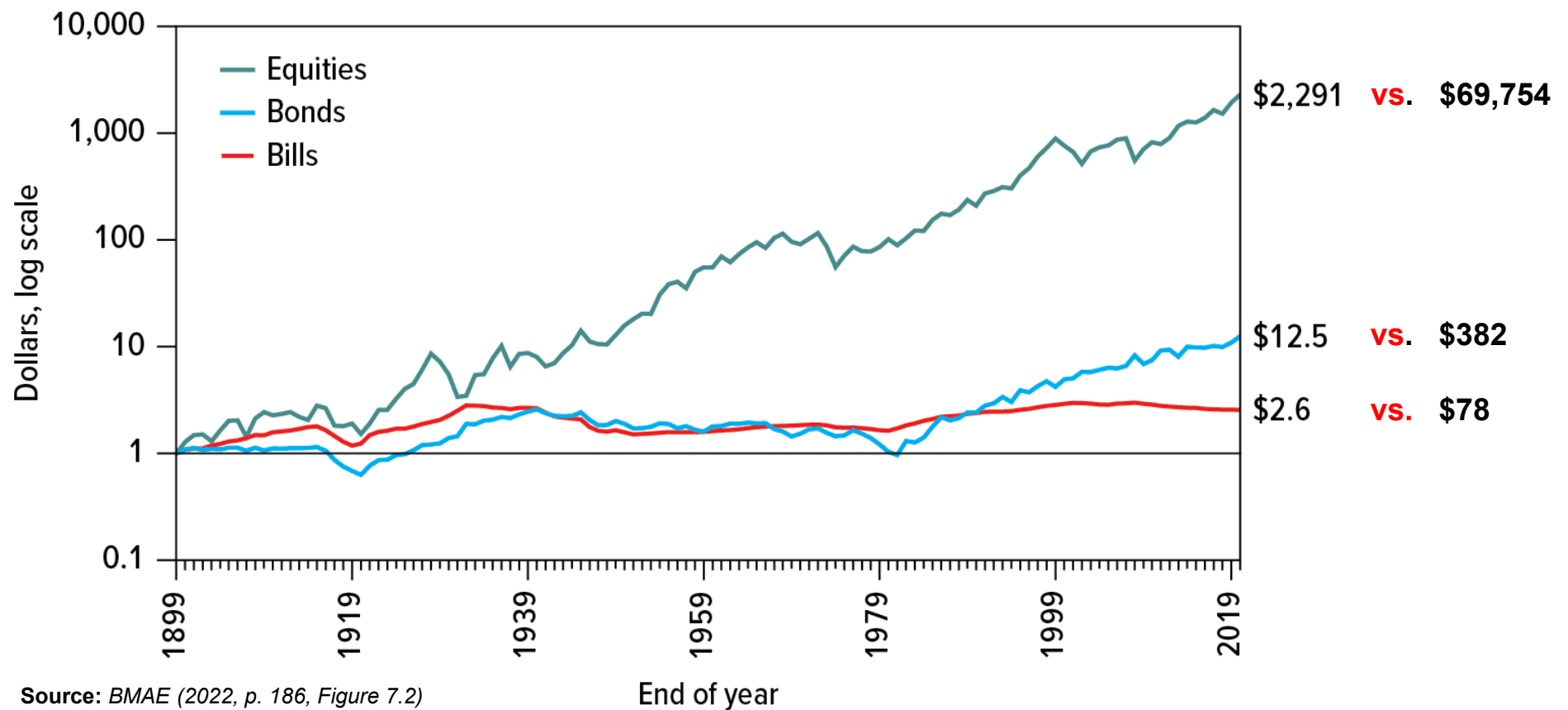
Different investments and their performance (II)

- The growth of a \$1 investment made in 1900 in one of three assets (assuming re-investment) until 2020



Different investments and their performance (III)

- The growth of a \$1 investment made in 1900 in one of three assets (assuming re-investment) until 2020 – inflation adjusted



Different investments and their performance (IV)

- While we focus on nominal values in this class, it is important to understand the power of inflation.
- For example, the average annual return of Treasury Bills was about 3.7%. The average inflation was about 2.8% on average.
=> Real return of T-Bills: 0.9% p.a.

Excursus – Inflation

Inflation and interest rates

- The **Consumer Price Index (CPI)** measures purchasing power by averaging the prices of goods and services in the consumption basket of an average urban family of four (as defined by the Bureau of Labor Statistics in the U.S.).
- The annual percentage change in CPI is used as a **measure of inflation** by both researchers and practitioners.
- The **classical theory of interest rates** was developed by Irving Fisher (1907).

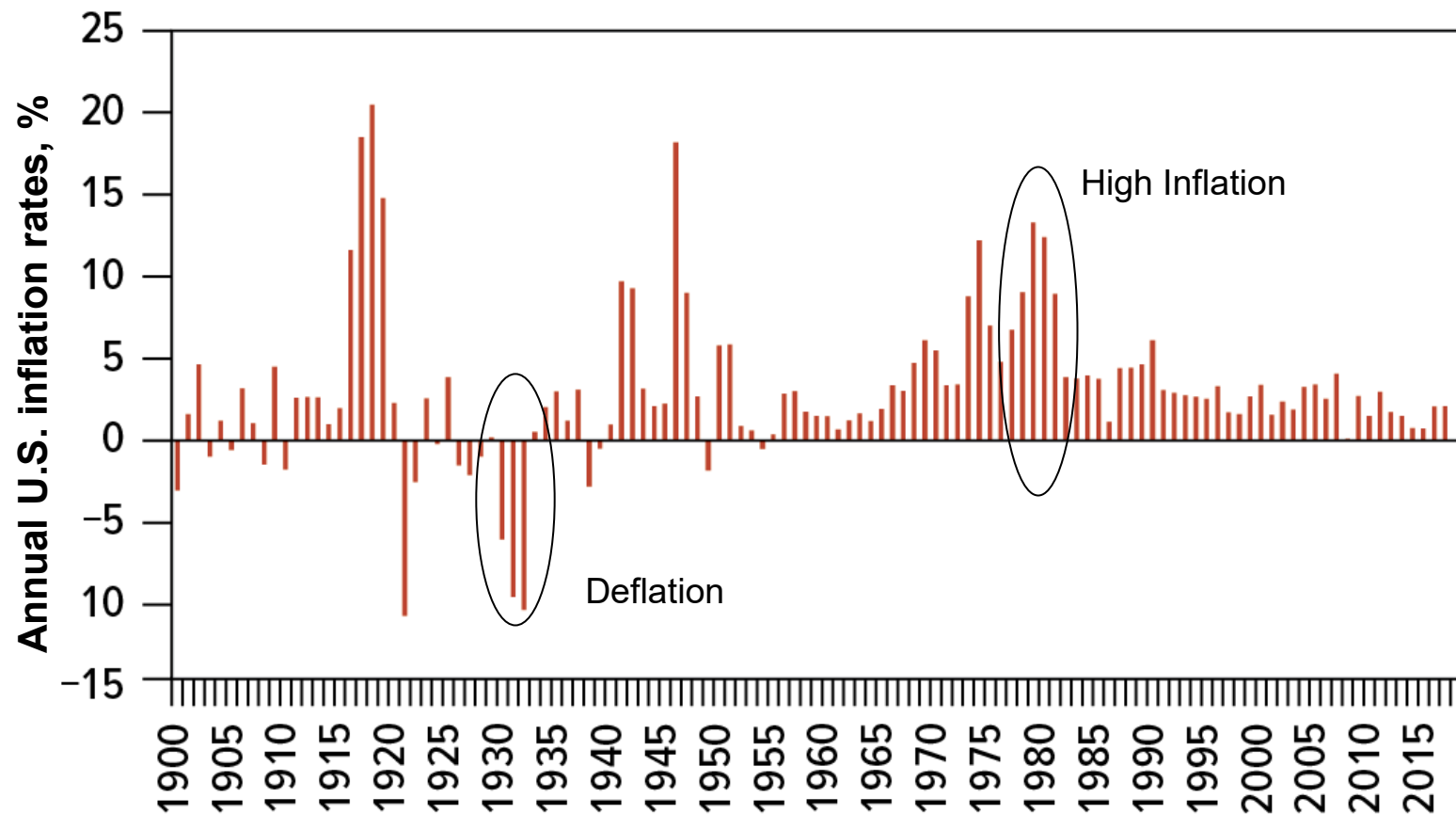
nominal r = real r $\times$ expected inflation (approximated)

$$1 + r_{\text{nominal}} = (1 + r_{\text{real}}) \cdot (1 + i)$$

Excursus – Inflation

Time variation

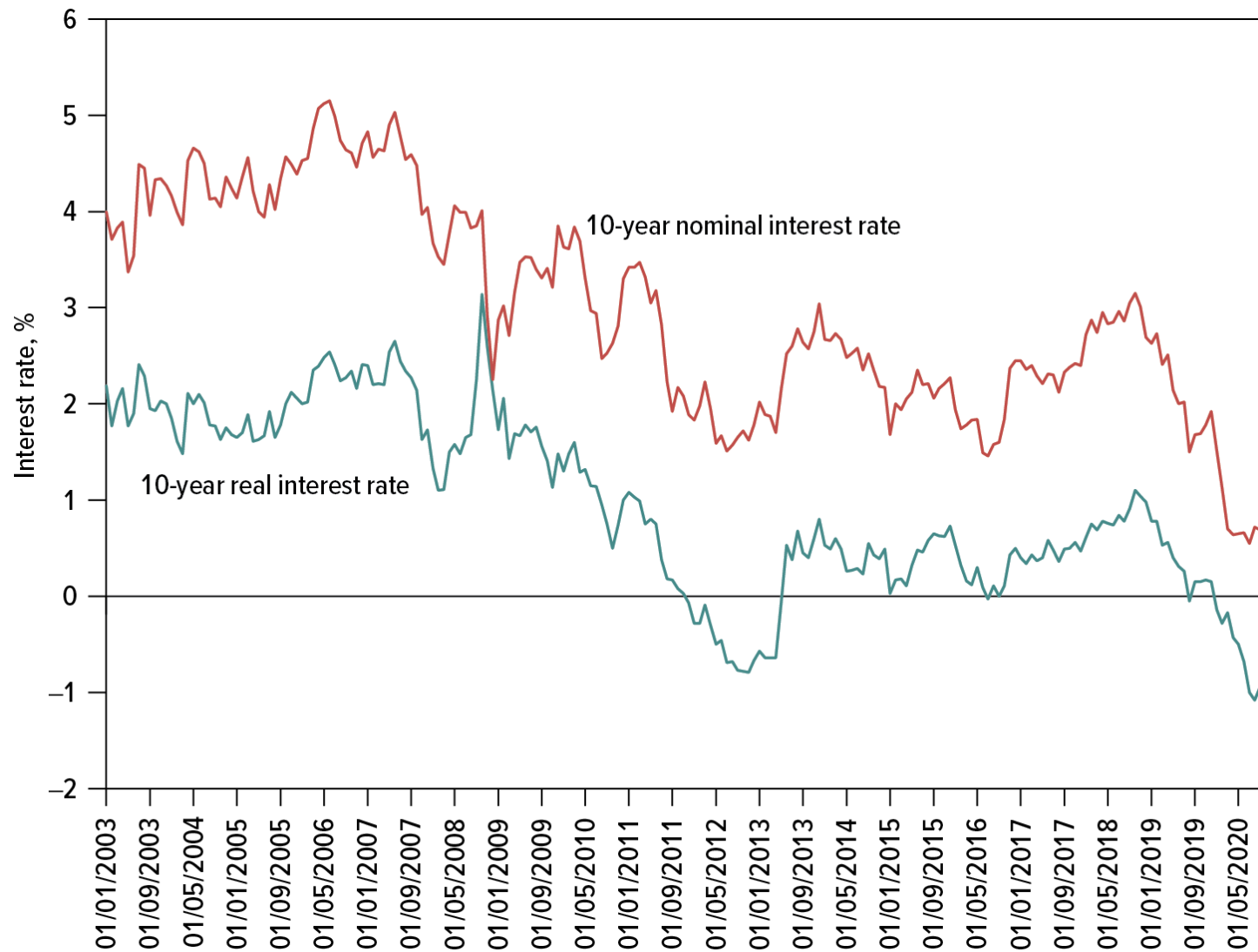
- Inflation measures the change of an economy's price level (goods and services).



Source: BMAE (2022, p. 70, Figure 3.5)

Excursus – Inflation

Real and nominal rates of interest



Source: BMAE (2022, p. 73, Figure 3.7)

Excursus – Inflation

Real and nominal rates of interest (II)

- The **real interest rate** is the **theoretical rate** you pay when you borrow money (or receive when you lend money).
 - The real interest rate cannot be observed; it can only be derived.
 - Real interest rates can be negative (due to low nominal interest rates and significant inflation).
 - The real interest rate is theoretically somewhat stable.
- Drivers of real interest rates:
 - Real interest rates are affected by the monetary policy of central banks.
 - People's willingness to save (supply of capital)
 - Opportunities for productive investments (demand side)

The equity risk premium

- Common stocks (avg. annual return: 11.5%) have offered an average **annual risk premium of 7.8%** when compared to the safest investment (U.S. Treasury Bills; avg. annual return: 3.7%).
- The equity risk premium is the risk premium that can be earned in the long-term from an investment in the stock market.
- Economically, it is the premium earned by investors willing to take on the stock market risk as compared to an investment in a risk-free asset.
- Example: If we use historical data, the premium depends on the time period we use (e.g., 1900-2007 produces quite a different premium than 1900-2009).

Empirical evidence on the equity risk premium

- Fernandez et al. (2019) analyze the equity risk premium for 2019:
 - Which equity risk premium do academics and professionals use for their analyses? How do they derive this risk premium?
 - They wrote an email (see excerpt below) to thousands of professors, analysts, and managers of companies in February 2019 and received approx. 1,800 replies

We are doing a **survey** about the **Market Risk Premium** (MRP or Equity Premium) and **Risk-Free Rate** that companies, analysts, regulators and professors use to calculate the **required return on equity** in different countries.

I would be grateful if you would kindly answer the following 2 questions. No companies, individuals or universities will be identified, and only aggregate data will be made public. I will send you the results in a month.

Best regards and thanks,

Pablo Fernandez, Professor of Finance, IESE Business School, Spain.

2 questions:

1. The Market Risk Premium that I am using in 2019

for USA is: _____ %

for _____ is: _____ %

for _____ is: _____ %

2. The Risk-Free rate that I am using in 2019

for USA is: _____ %

for _____ is: _____ %

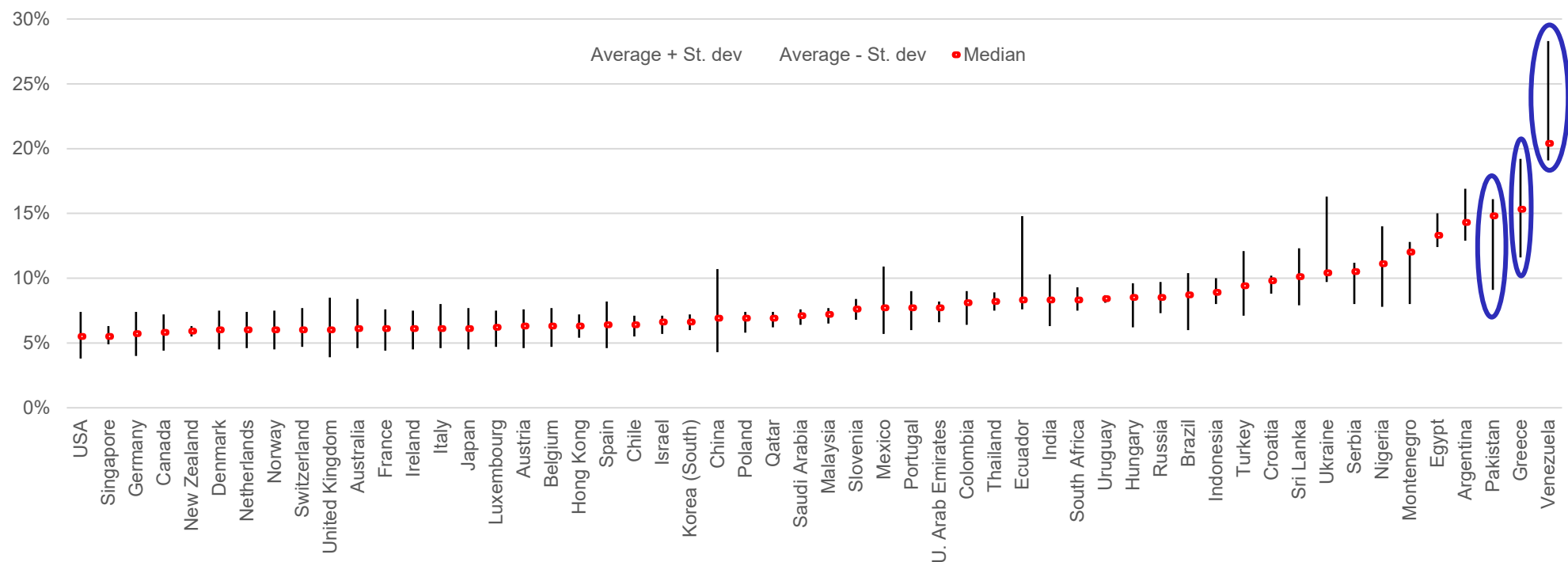
for _____ is: _____ %

Source: Fernandez, Martinez, and Fernandez Acin, 2019, Market Risk Premium and Risk-Free Rate used for 69 countries in 2019: a survey.



Empirical evidence on the equity risk premium (II)

- Fernandez et al. (2019) report the median equity risk premium as well as the dispersion of the answers.

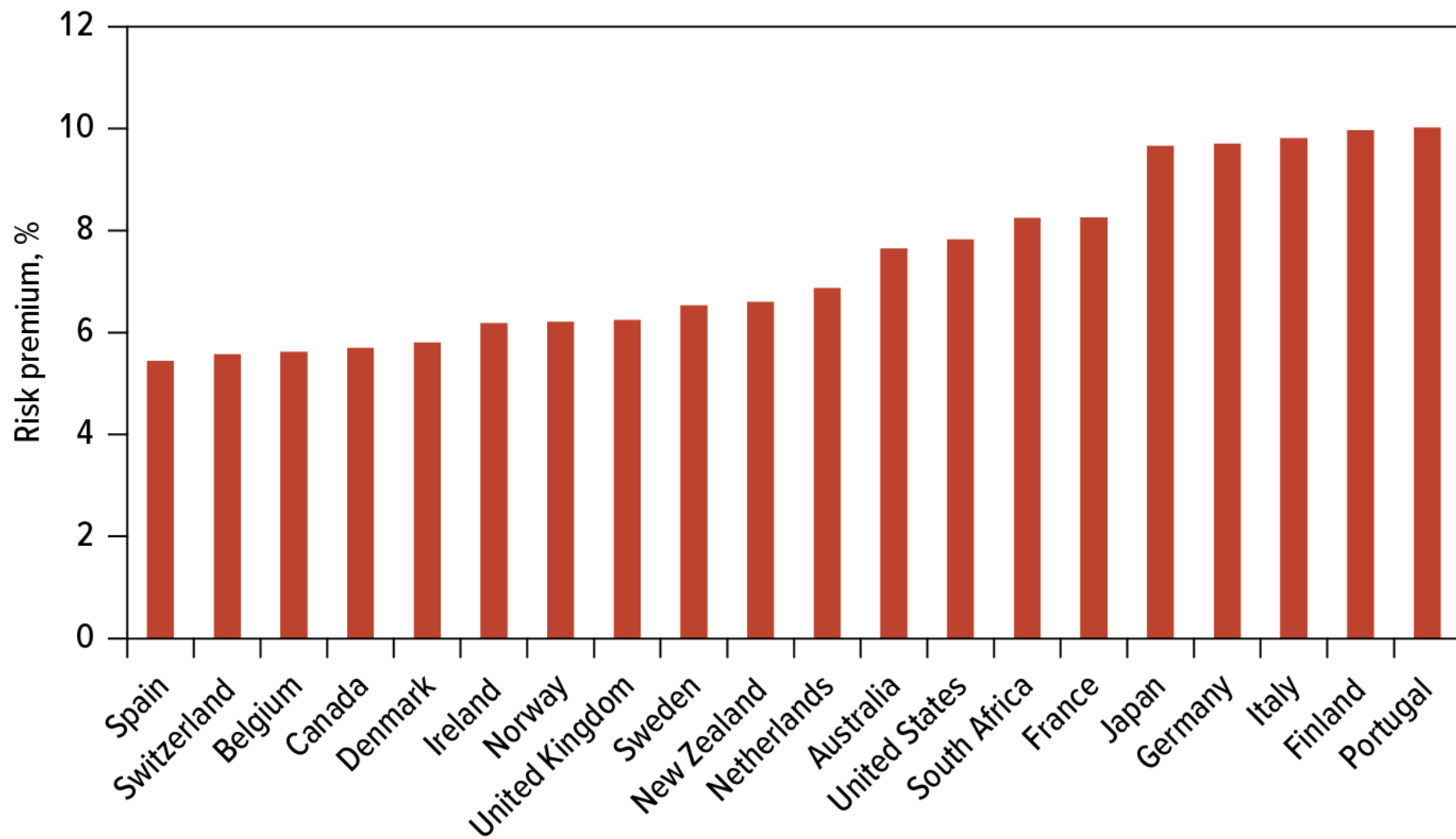


Source: Fernandez, Martinez, and Fernandez Acin, 2019, Market Risk Premium and Risk-Free Rate used for 69 countries in 2019: a survey.

Empirical evidence on the equity risk premium (III)

- Where do estimates of the risk premium come from?
 - Damodaran
 - Historical data
 - Ibbotson/Morningstar
 - Brealey, Myers, and Allen and other books
 - Bloomberg / Analyst forecasts
 - Fernandez et al. surveys
 - ...

Empirically estimated equity risk premium across countries, 1900-2020



Source: BMAE (2022, p. 187, Figure 7.3)

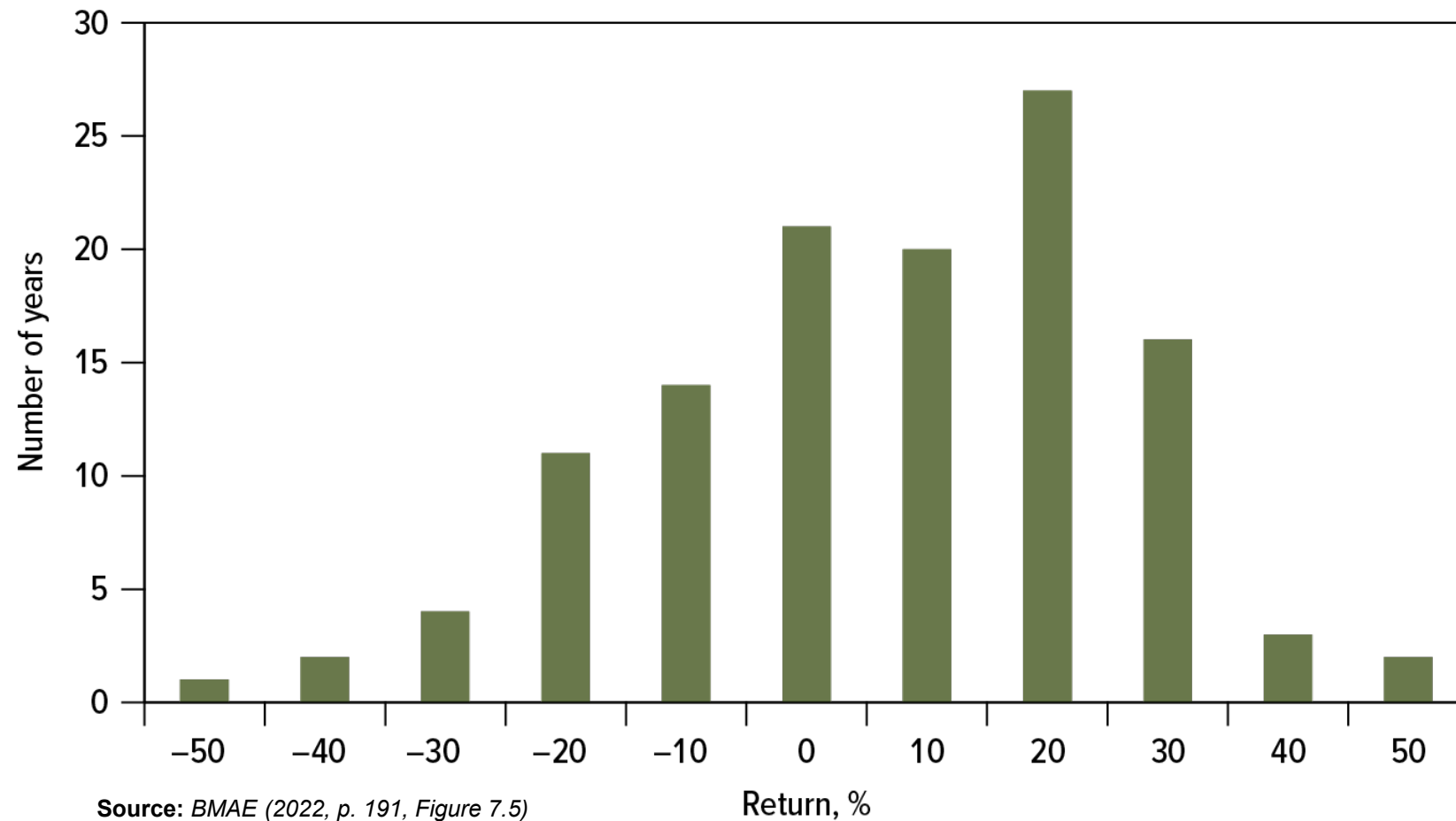
The equity risk premium (II)

- What drives the equity risk premium? And why does it differ so much across countries?
 - Higher risk in some countries (e.g., Italy)
 - Higher inflation in some countries (these are nominal values)
 - Ex-post some countries might have been more fortunate than could have been expected
 - Valuation levels increased (e.g., based on dividend yield or M/B-ratio) which could indicate optimism and result in a lower equity risk premium in the future.

Measuring risk: The standard deviation

- There is an important difference between the **expectation** and the **actual realization** in any given year.
- The realized stock returns are by no means equal to or close to their historical average value but vary a lot across years (and months, weeks, days...).
- In order to measure this “spread” of returns over the years we can compute the **variance** or **standard deviation** of the returns.
- Besides computing the average annual return (for U.S. stocks from 1900-2020: 11.5%) we can compute the expected/realized squared deviation from this average annual return, and use this measure to compare different portfolios regarding their level of risk.

Histogram of annual returns for the U.S., 1900-2020



Measuring risk: The standard deviation (II)

- The variance is defined as follows:

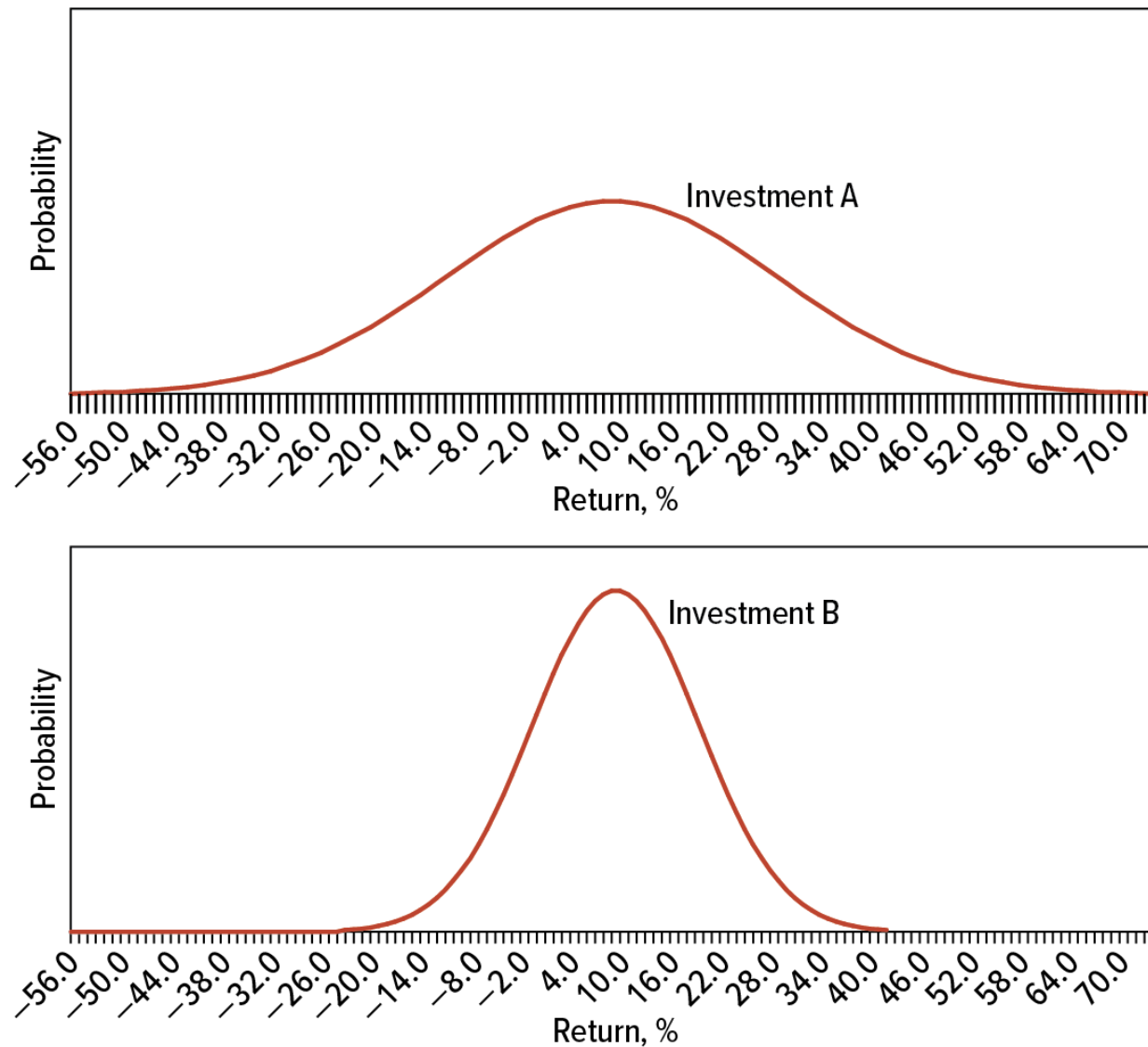
$$\text{Variance}(\tilde{r}_m) = \frac{1}{N-1} \sum_{t=1}^N (\tilde{r}_{mt} - r_m)^2$$

where $\tilde{r}_{mt}$ is the market return in period t and r_m is the mean of the values of $\tilde{r}_{mt}$.

- The standard deviation, or volatility, is the square root of the variance:

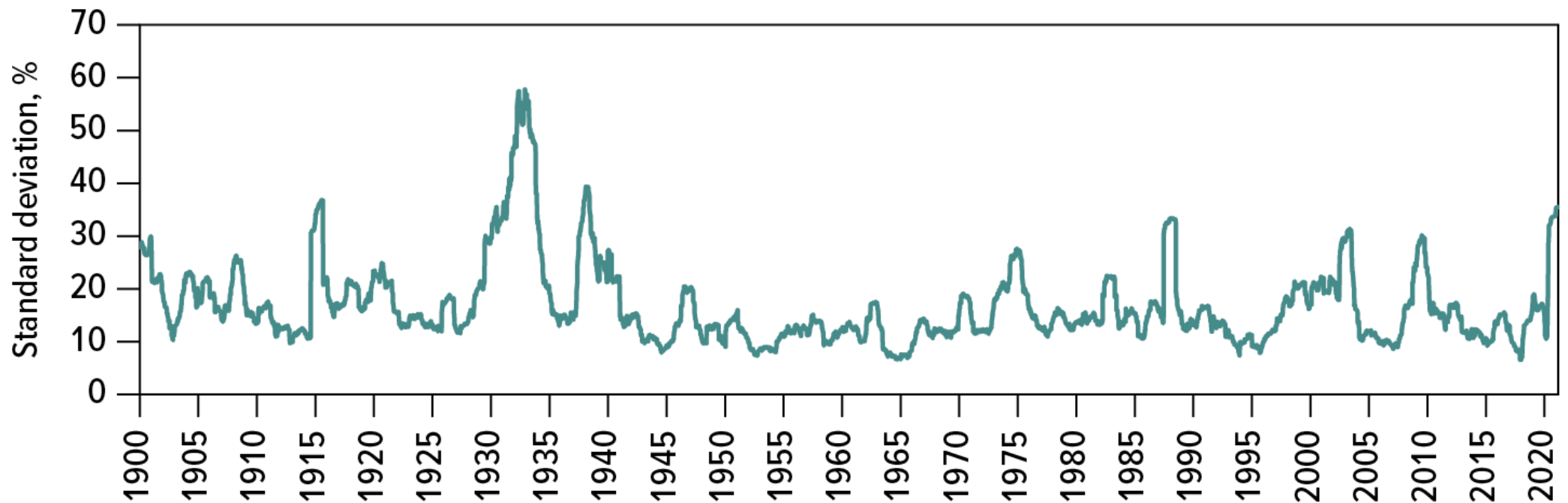
$$\text{Standard deviation of } \tilde{r}_{mt} = \sqrt{\text{Variance}(\tilde{r}_m)}$$

Which investment is riskier?



Source: BMAE (2022, p. 192, Figure 7.7)

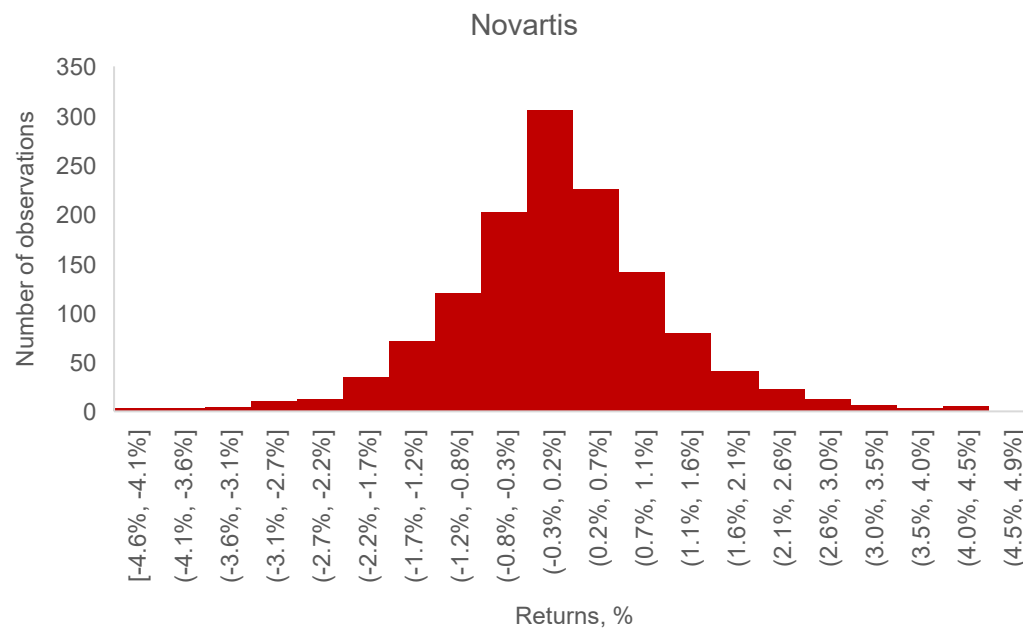
Annualized standard deviation of the Dow Jones Industrial Average, 1900-2020



Source: BMAE (2022, p. 194, Figure 7.9)

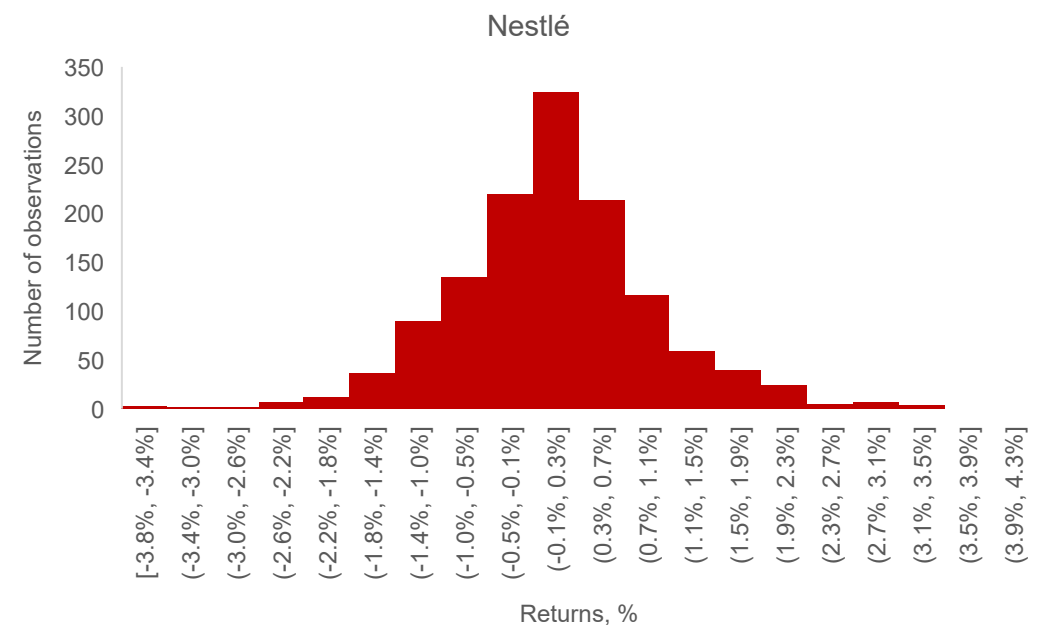
What is the standard deviation of Novartis and Nestlé?

Novartis (Jan 2014 – Dec 2018)



Standard Deviation: 18.51%

Nestlé (Jan 2014 – Dec 2018)



Standard Deviation: 14.96%

Data source: Datastream

Standard deviations for selected U.S. stocks, Mar 2015 – Feb. 2020 (in % per year)

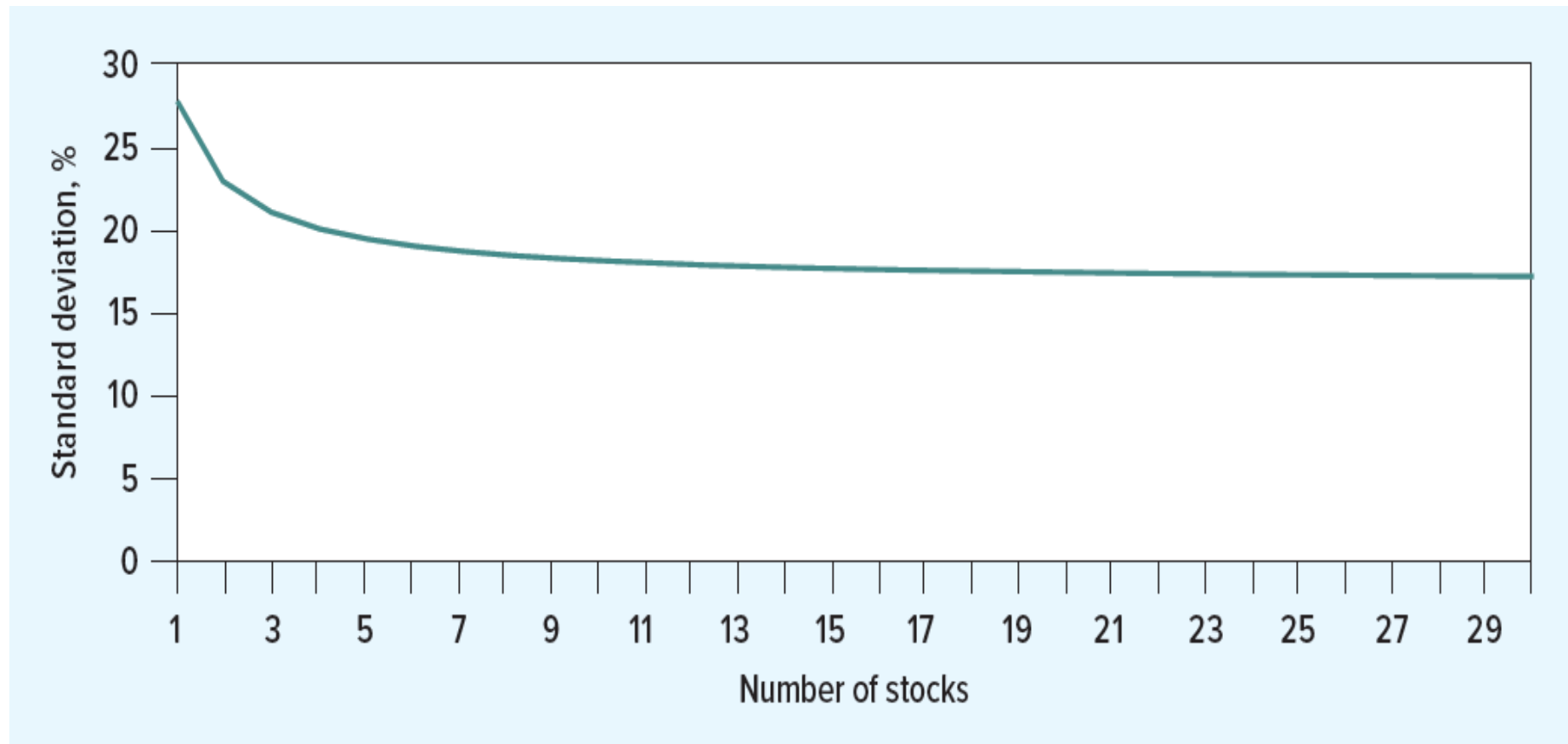
Stock	Standard Deviation σ	Stock	Standard Deviation σ
United States Steel	76.4%	Wells Fargo	21.6%
Tesla	48.1	ExxonMobil	19.4
Newmont	36.7	Consolidated Edison	16.5
Southwest Airlines	30.5	Johnson & Johnson	14.4
Amazon	28.3	Coca-Cola	12.6

Source: BMAE (2022, p. 195, Table 7.3)

Reducing risk through diversification

- The standard deviations for the individual U.S. stocks on the last slide are quite high as compared to the standard deviation of the market portfolio (12.0%).
- Where does the difference come from? **Diversification benefits.**
- The intuition is simple: If you invest all your money into one stock, your wealth will entirely depend on the **specific (or idiosyncratic) risk** of this one company (i.e., “putting all eggs in one basket”).
- However, if you split your wealth and invest 1/100 into 100 different companies, you diversify away the specific risks the individual companies face.
- For instance, a spike in oil prices has a huge impact on airline companies, but has minor implications for large food companies.

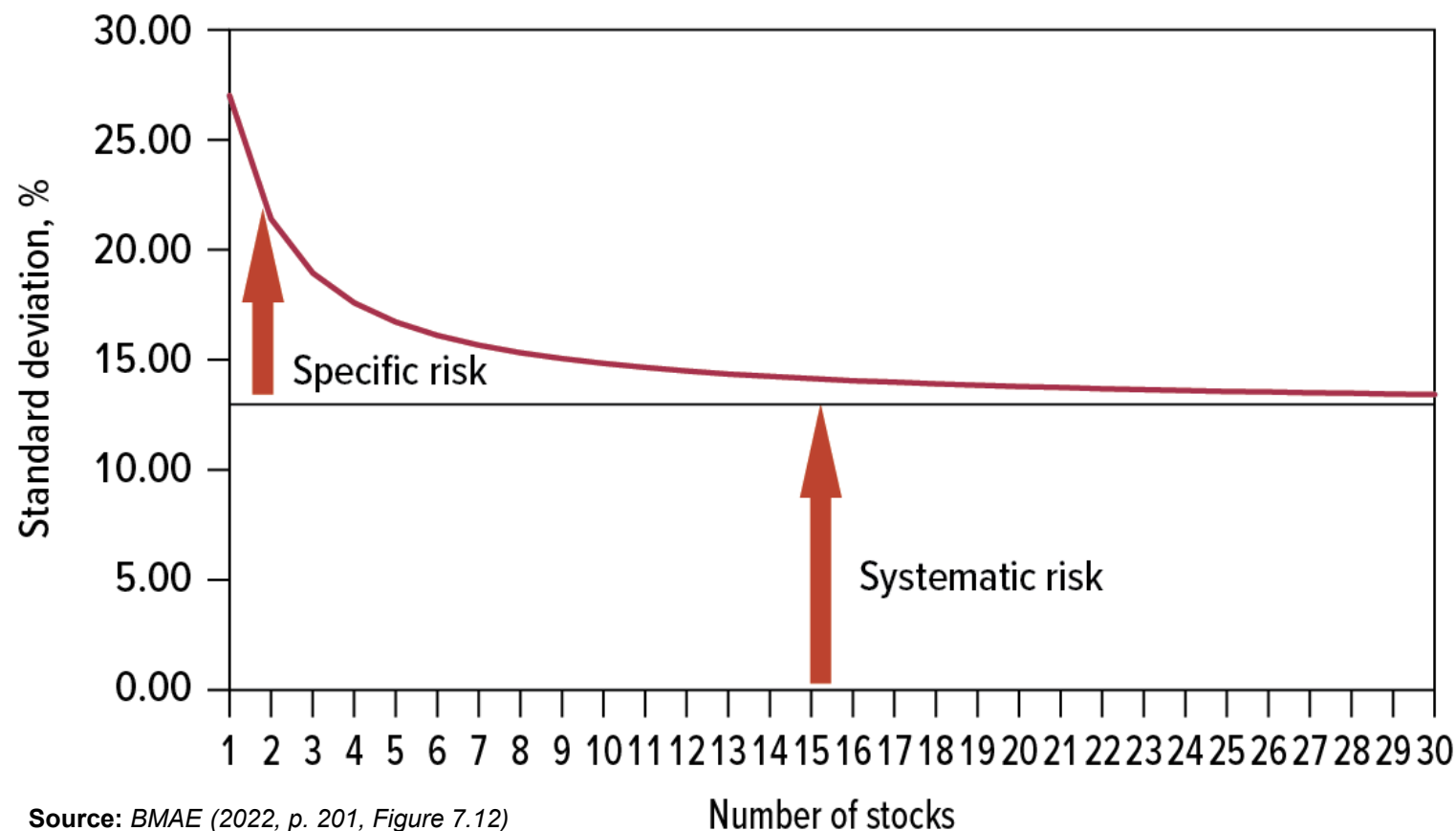
Reducing risk through diversification: Adding stocks to a portfolio



Source: BMA (2019, p. 179, Figure 7.9)

Reducing risk through diversification: Specific vs. market risk

- The only risk left (which you cannot diversify away) is the **market** (or **systematic**) risk.



Source: BMAE (2022, p. 201, Figure 7.12)

Correlation drives the portfolio benefits

- The **covariance** is a measure of how much two random variables change together. If the greater values of one variable mainly correspond with the greater values of the other variable, and the same holds for the smaller values, i.e., the variables tend to show similar behavior, the covariance is a positive number. If the variables show opposite behavior, the covariance is a negative number.
- The **correlation coefficient** measures the strength and direction of the linear relationship between two variables. It is a standardized version of the covariance and is bound to a $(-1, 1)$ interval.

Correlation drives the portfolio benefits (II)

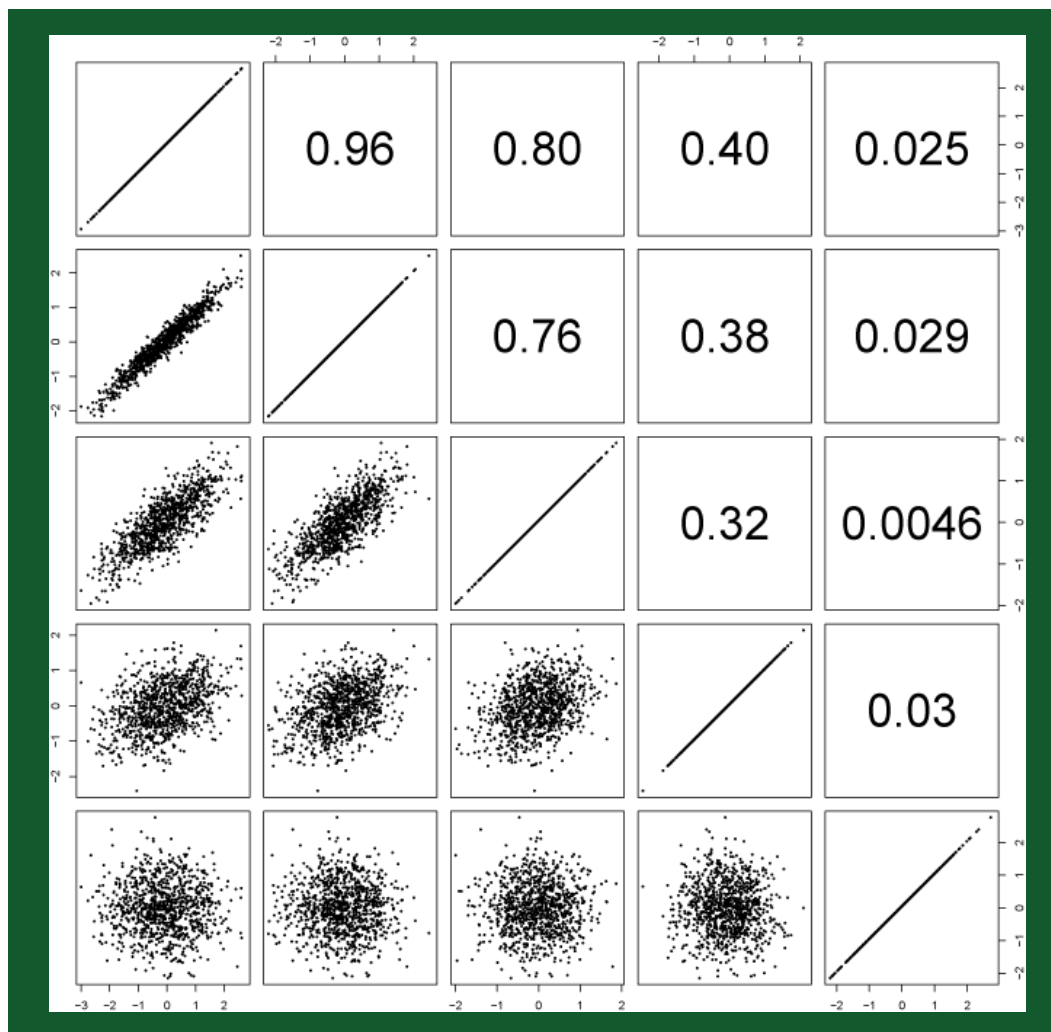
- The magnitude of the covariance is not easy to interpret. Its normalized version, the correlation coefficient, however, shows by its magnitude the strength of the linear relation.
- The covariance is defined as:

$$\text{Cov}[r_A, r_B] = \frac{1}{N-1} \cdot \sum_{t=1}^N (r_{A,t} - \mu_A) \cdot (r_{B,t} - \mu_B)$$

- The correlation coefficient is defined as:

$$\text{Corr}_{A,B} = \rho_{A,B} = \frac{\text{Cov}[r_A, r_B]}{SD[r_A] \cdot SD[r_B]}$$

Correlation coefficient: Illustration



1'000 pairs of normally distributed numbers are plotted against one another in each panel (bottom left of the graph), and the corresponding correlation coefficients are reported (top right of the graph). Along the diagonal, each set of numbers is plotted against itself, defining a line with a correlation of +1.

Calculating portfolio returns and portfolio risk

- Calculating the **expected return of a portfolio of stocks** is simple: It is the weighted average of the expected returns of the individual securities in the portfolio.
- The **portfolio risk** (variance) is the sum of all individual **variances** multiplied by their weights squared and all **covariances** multiplied by the weights of both respective stocks.
- A simple **example** of a 2-stock portfolio helps to understand the concept:

Calculating portfolio risk with two securities

	Stock 1 weight: x_1 risk: σ_1	Stock 2 weight: x_2 risk: σ_2
Stock 1 weight: x_1 risk: σ_1	$x_1 \cdot \sigma_1 \cdot x_1 \cdot \sigma_1$ $= x_1^2 \cdot \sigma_1^2$	$x_1 \cdot x_2 \cdot \sigma_{21}$ $= x_1 \cdot x_2 \cdot \rho_{12} \cdot \sigma_1 \cdot \sigma_2$
Stock 2 weight: x_2 risk: σ_2	$x_2 \cdot x_1 \cdot \sigma_{12}$ $= x_1 \cdot x_2 \cdot \rho_{12} \cdot \sigma_1 \cdot \sigma_2$	$x_2 \cdot \sigma_2 \cdot x_2 \cdot \sigma_2$ $= x_2^2 \cdot \sigma_2^2$

Calculating portfolio returns and portfolio risk (II)

- In reality, a portfolio has more than 2 stocks. But the concept remains the same!

1. Compute all individual variances
2. Multiply all individual variances with their weights squared
3. Compute the pairwise covariances of all stocks
4. Multiply by the respective weights
5. Sum up all weighted variances and weighted covariances

You now understand how to compute and compare the returns and risk of different portfolios!

Limits to diversification

- If we have a portfolio with N shares and the proportion of $1/N$ is invested in each share:

$$\text{Portfolio variance} = \frac{1}{N} * \text{average variance} + \left(1 - \frac{1}{N}\right) * \text{average covariance}$$

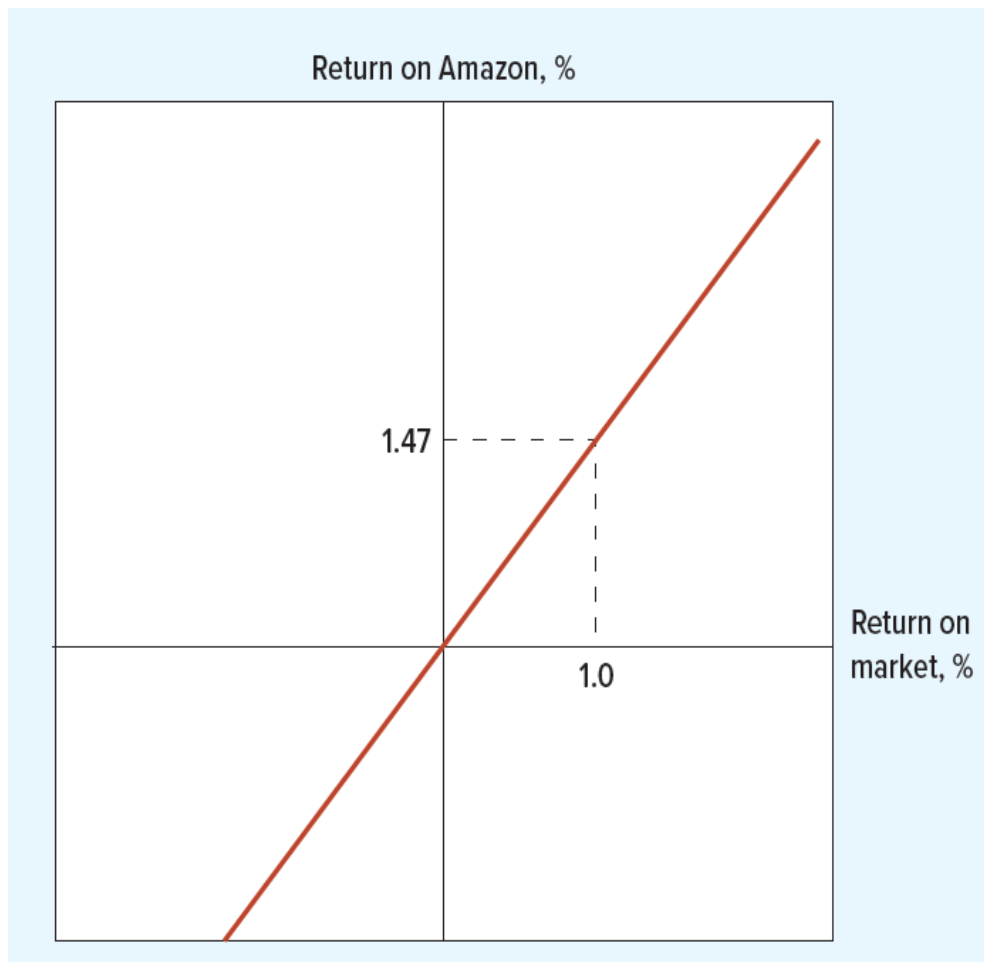
- Hence, as N increases, the portfolio variance converges to the average covariance.
- If the average covariance is equal to zero, it would be possible to obtain a risk-free portfolio.

Beta and portfolio risk

- Once you have a diversified portfolio (i.e., no specific risk), all risk in the portfolio is **market risk**.
- But different stocks react differently to market risk, i.e., show varying degrees of sensitivity to market movements (e.g., utility companies react in general less to market movements than technology companies).
- The sensitivity of the stock to market movements is called **beta (β)**: Market beta is a measure of market risk of the respective stock.
- Statistically, beta is the regression coefficient from a regression of the returns of a specific stock on the returns of the market portfolio.

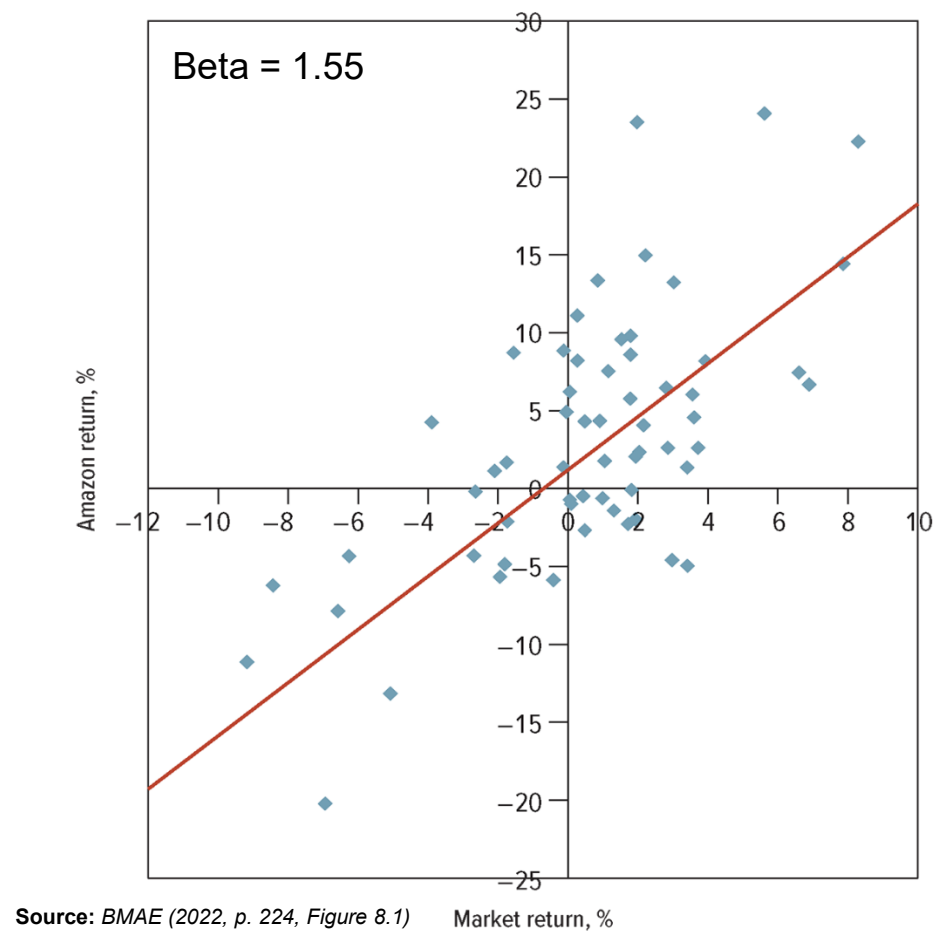
Beta and portfolio risk: Example

Amazon, beta (2013-2017)



Source: BMA (2019, p. 187, Figure 7.14)

Amazon, beta (2015-2020)



Source: BMAE (2022, p. 224, Figure 8.1)

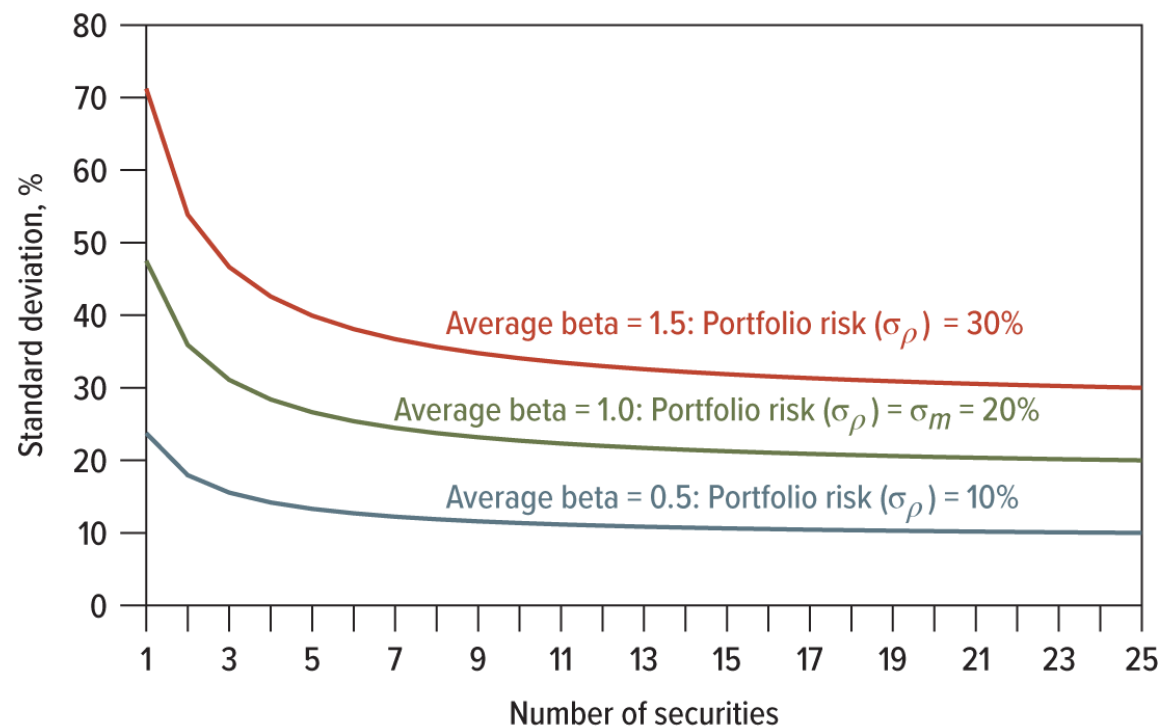
Beta: A measure of systematic risk

- Beta is the **risk contribution** of an asset to a diversified portfolio (more precisely to the market portfolio).
- As beta is simply the slope coefficient of a (univariate) regression of R_i on R_m , it is defined as:

$$\beta_i = \frac{Cov(R_i, R_m)}{Var(R_m)} = \frac{\sigma_{im}}{\sigma_m^2}$$

Betas determine portfolio risk

- The beta of a portfolio is the weighted average of the betas of the individual stocks within the portfolio:



Source: BMAE (2022, p. 227, Figure 8.2)

- The beta of a well-diversified portfolio also defines its total risk.

How to calculate variances, covariances, and betas

- Excel example based on six monthly observations:

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Month	Market return	Novartis return	Nestlé return	Deviation from average market return	Deviation from average Novartis return	Deviation from average Nestlé return	Squared deviation from average market return	Product of deviations from average returns Novartis (col 5 x 6)	Product of deviations from average returns Nestlé (col 5 x 7)
1	-2.54	-2.61	2.40	-2.94	-5.10	3.51	8.66	15.02	-10.32
2	-1.03	-1.20	-2.97	-1.43	-3.70	-1.86	2.05	5.30	2.67
3	7.56	12.79	4.69	7.16	10.30	5.80	51.26	73.72	41.50
4	-1.86	-2.67	0.02	-2.26	-5.16	1.13	5.12	11.67	-2.55
5	1.48	4.91	3.49	1.08	2.42	4.59	1.17	2.61	4.96
6	-1.20	3.74	-14.26	-1.60	1.25	-13.16	2.57	-2.00	21.09
Average	0.40	2.49	-1.10			Total	70.82	106.32	57.34

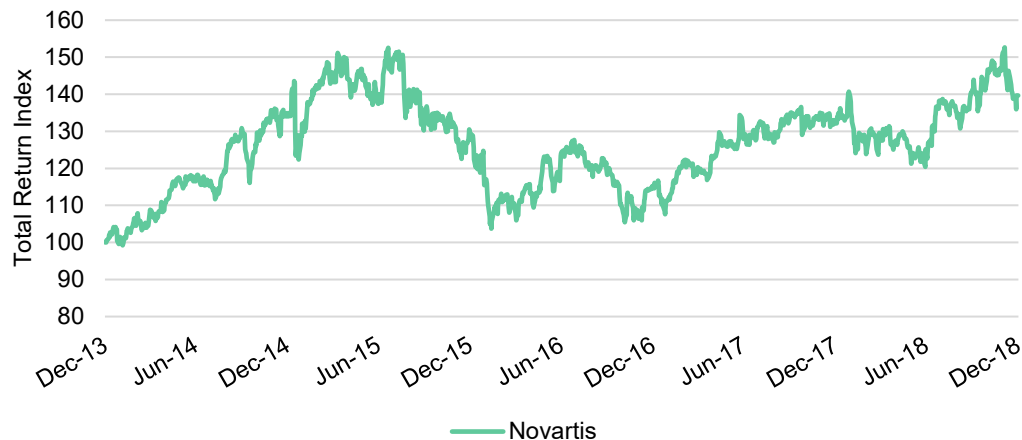
* monthly returns in % from June 2018 to November 2018

Variance:	$\sigma_m^2 = 70.82 / 5 = 14.16$
Covariances:	$\sigma_{\text{Novartis},m} = 106.32 / 5 = 21.26$ $\sigma_{\text{Nestlé},m} = 57.34 / 5 = 11.47$
Betas:	$\beta_{\text{Novartis}} = \sigma_{\text{Novartis},m} / \sigma_m^2 = 21.26 / 14.16 = 1.50$ $\beta_{\text{Nestlé}} = \sigma_{\text{Nestlé},m} / \sigma_m^2 = 11.47 / 14.16 = 0.81$

So, which of the two companies is more risky?

- Let's re-visit the introductory example:

Novartis (Jan 2014 – Dec 2018)



Data source: Datastream

Beta: 1.187

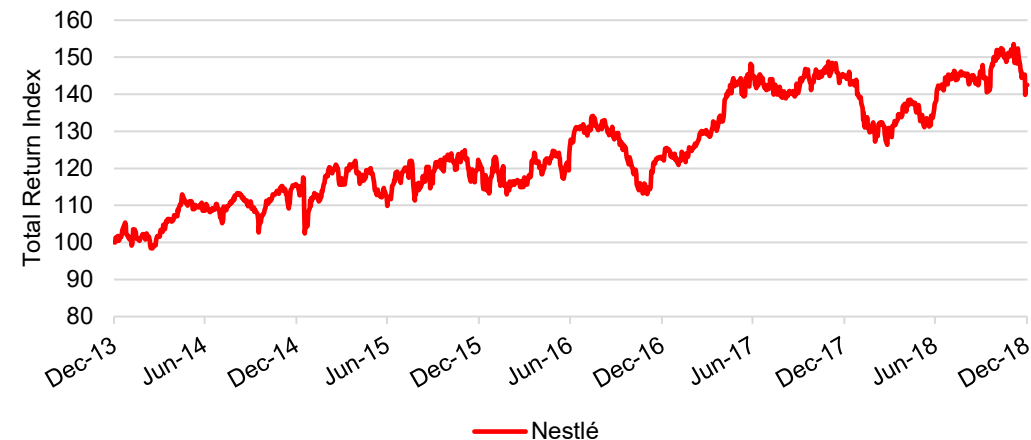
Standard Deviation: 18.51%

Arithmetic Mean Return: 7.67%

Geometric Mean Return: 6.67%

OR

Nestlé (Jan 2014 – Dec 2018)



Beta: 0.752

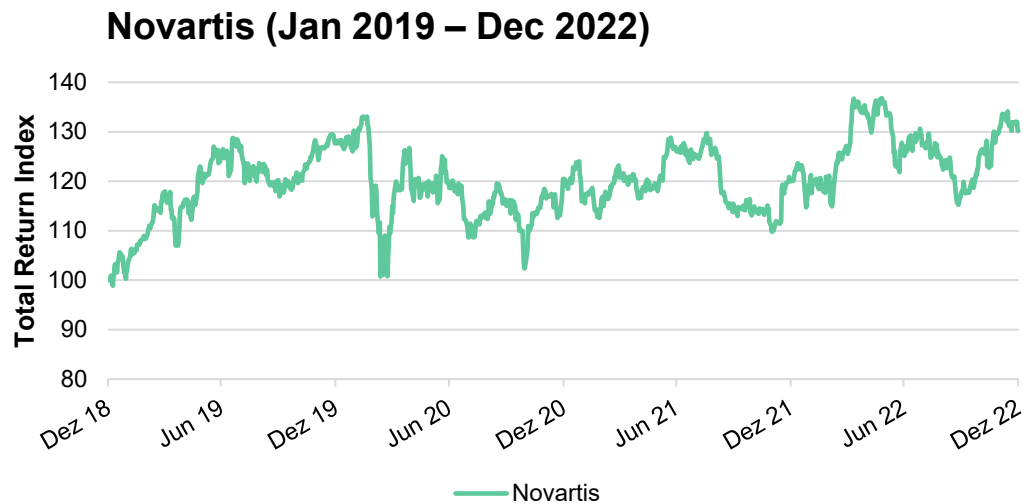
Standard Deviation: 14.96%

Arithmetic Mean Return: 8.21%

Geometric Mean Return: 7.08%

Which of the two companies is more risky? – Update

- What about the subsequent four years (2019-2022)?:



Data source: Datastream

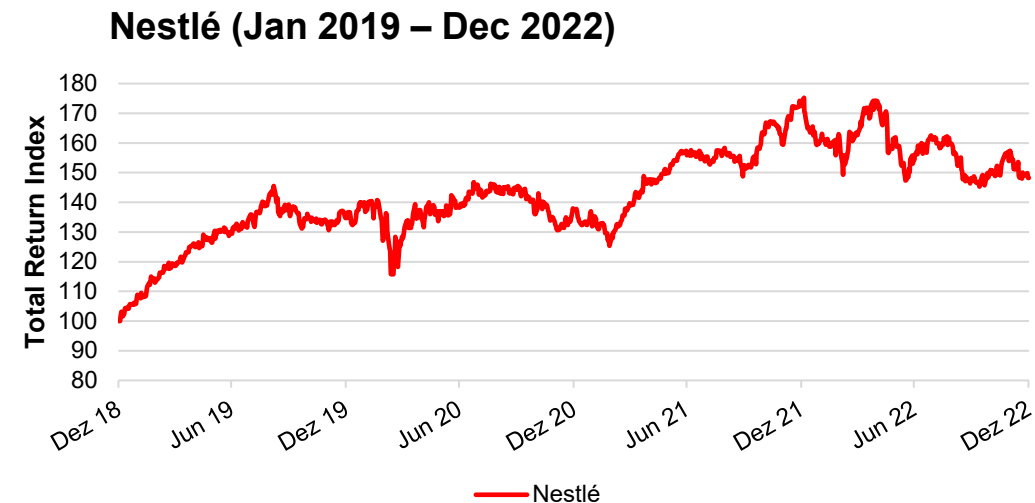
Beta: 1.014

Standard Deviation: 18.51%

Arithmetic Mean Return: 8.35%

Geometric Mean Return: 6.81%

OR



Beta: 0.771

Standard Deviation: 16.55%

Arithmetic Mean Return: 11.39%

Geometric Mean Return: 10.34%

Key takeaways

1. Different investments show different degrees of risk, because they have different exposure to **idiosyncratic (specific) risks** and **market risk**.
2. In finance, we measure **total risk** by the variance (standard deviation) from the expected returns.
3. Diversification reduces risk because it eliminates the specific risk of individual securities when combined in a portfolio of stocks.
4. The only risk that cannot be diversified away is the **market risk**.
5. Beta (β) measures the sensitivity of an individual stock or portfolio of stocks to market movements.

Key takeaways (II)

6. Since the risk of a well-diversified portfolio only depends on market risk, beta is the appropriate measure to
- (1) describe the level of risk of an investment (e.g., stock, mutual fund, etc.), and
 - (2) compare different assets regarding their expected returns and beta exposure as a measure of risk contribution to the portfolio, which are the two dimensions of the Capital Asset Pricing Model (CAPM, next lecture).

Literature

Mandatory readings

- Brealey, R., Myers, S., Allen, F., and A. Edmans, 2022, Principles of Corporate Finance, 14th Edition, New York: McGraw-Hill, Chapter 7 (Introduction to Risk, Diversification and Portfolio Selection).

Additional readings

- Elton, E.J., Gruber, M.J., Brown, S.J., and W.N. Goetzmann, 2017, Modern Portfolio Theory and Investment Analysis, 9th edition, John Wiley & Sons.
- Fernandez, P., Martinez, M., and I. Fernandez Acin, 2019, Market Risk Premium and Risk-Free Rate used for 69 countries in 2019: a survey, Working Paper, IESE Business School.